## Atmospheric correction

Launcher: `Toolbox/atmcorr.bat`

### Purpose

Atmospheric correction converts sensor digital numbers into estimated bottom-of-atmosphere surface reflectance. The current GUI applies DOS3 (Dark Object Subtraction) to both the multispectral and panchromatic products in a WorldView product folder. This should normally be the first processing step.

DOS3 estimates haze from dark pixels, applies sensor radiometric calibration and acquisition geometry, compensates approximately for Rayleigh scattering, and clips negative reflectance to zero. It is an empirical correction, not a full atmospheric radiative-transfer model.

### Inputs and parameters

| Option | What to provide | Behaviour |
|---|---|---|
| **Input Image Folder** | The top-level directory of one WV3 or LEGION-06 product | It must contain one folder ending `_MUL` and one ending `_PAN`. Those folders must retain their `.TIF`, `.TIL`, `.IMD`, and `.XML` product metadata; the parent product folder must retain its `*README.TXT`. The first matching MUL and PAN folders are used. |
| **Output Folder** | An existing folder, or blank | Blank creates `<input-folder>/<input-folder-name>_BOA`. If that automatic folder already exists, the GUI asks you to choose another folder or rerun. A manually entered folder is expected to exist. |

| Option | What to provide | Behaviour |
|---|---|---|
| **Vector layer** | Optional clip polygon | The raw product is cropped before correction. Pixels outside the cutline become NoData `−9999`. Leave blank to process the full scenes. |
| **Sensor** | `WV3` or `LEGION_06` | Select the sensor that produced the imagery. It controls calibration coefficients and metadata interpretation. Default: `WV3`. |
| **Atmospheric Correction formula** | `DOS3` | DOS3 is the currently available formula and the default. |
| **Scale** | A non-negative integer | `0` keeps floating-point reflectance, normally around `0`–`1`. A positive value multiplies reflectance and writes signed 16-bit integers. `10000` is recommended for the conventional Bayesian pansharpening workflow. Negative and non-integer values are rejected. |
| **Output compression** | `Lossless` or `Near-lossless` | Lossless (default) is bit-exact. Near-lossless (LERC) allows an error of at most 5e-5 per pixel and only applies with **Scale** `0`; scaled integer outputs are always lossless. Keep the output lossless if it feeds pansharpening or another tool: see Output compression. |

### Outputs

For each detected MUL and PAN image, the tool creates:

- `<source-name>_DOS3.tif`

Outputs are Cloud Optimized GeoTIFFs with NoData `-9999` and overviews: DEFLATE (lossless), or LERC (near-lossless, Float32 only). The source data are not modified. Temporary clipped files are removed after a successful run.

### Choosing a scale

- Use `10000` when the next step is conventional Bayesian pansharpening. It preserves four decimal places while giving the solver image-like numeric magnitudes.
- Use `0` when another scientific workflow specifically expects floating-point reflectance. If Bayesian pansharpening follows, start with a much lower lambda (around `0.1`) and inspect band statistics carefully.
- Because positive scaling writes `int16`, values above about 3.2767 at scale 10000 would clip to the data-type maximum. Normal reflectance usually stays below this limit.

### Quality checks

Open both outputs and verify their CRS, footprint, band count, and NoData border. Inspect per-band histograms: most valid reflectance should be non-negative, and multispectral bands should retain distinct distributions. Strong haze, cloud, water, shadow, snow, or an unrepresentative dark-pixel estimate can bias DOS3.